

Praween Siritanasak

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📍 Chiangmai, Thailand

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Education

- Ph.D. University of California, San Diego**, Physics — Experimental Cosmology San Diego, CA, USA
2009 – 2018
- Dissertation: "Precise Measurement of B-mode Polarization Signal from the Cosmic Microwave Background with the POLARBEAR and the Simons Array"
 - Awarded Royal Thai Government Scholarship (2010–2017)
- M.Sc. University of California, San Diego**, Physics San Diego, CA, USA
2009 – 2012
- B.Sc. Mahidol University**, Physics Bangkok, Thailand
2005 – 2009
- First-class Honours
 - Sritrangthong Scholarship (2005–2009)

Experience

- National Astronomical Research Institute of Thailand (NARIT)**, Researcher Chiangmai, Thailand
Oct 2020 – present
5 years 10 months
- CMB polarization instrumentation and detector development
 - Member of the POLARBEAR/Simons Array collaboration
- Center for Astrophysics and Space Sciences (CASS), UC San Diego**, Postdoctoral Research Fellow San Diego, CA, USA
June 2018 – Oct 2020
2 years 5 months
- Development of next-generation CMB receiver systems under Professor Brian Keating
 - Testing and characterization of transition-edge sensor (TES) bolometers
- Department of Physics, UC San Diego**, Graduate Research Assistant San Diego, CA, USA
2009 – 2018
9 years
- The POLARBEAR and the Simons Array CMB polarization experiment

Awards

- Royal Thai Government Scholarship** 2010
- Full-ride graduate scholarship for doctoral study abroad, 2010–2017
- Sritrangthong Scholarship** 2005
- Merit-based undergraduate scholarship, Mahidol University, 2005–2009

Skills

- CMB Instrumentation:** POLARBEAR, Simons Array, TES detectors, receiver development, cryogenics, anti-reflection coatings
- Data Analysis:** CMB polarization analysis, B-mode detection, gravitational lensing, power spectrum estimation
- Programming & Tools:** Python, C, MATLAB, Mathematica, LaTeX, Solidworks, HFSS
- Operating Systems:** macOS, Linux (RedHat, Ubuntu, Linux Mint)

Languages

- Thai:** Native
- English:** Fluent

Presentations

- **Oral:** "The Current Status of POLARBEAR and Simons Array Experiment," 19th Siam Physics Congress, Ayutthaya, Thailand (2024)
- **Oral:** "Instrument Development of the CMB Polarization POLARBEAR-2 Experiment," 230th American Astronomical Society Meeting, Austin, Texas, USA (2017)
- **Poster:** "The Broadband Anti-Reflection Coated Extended Hemispherical Silicon Lenses for POLARBEAR-2 Experiment," 16th International Workshop on Low Temperature Detectors, Grenoble, France (2015)